

Homework 09 Two Way ANOVA / ANCOVA / GLM

Due by 11:59pm, Tuesday, 4.21.26

S&DS 2300/5300/ENV 757

This assignment uses data from the International Social Survey Program on Environment from 2020. There are 224 questions from 21,718 individuals across 14 countries.

The data you'll need is [here](#). Be aware that it will take a few moments to load this data.

You'll also want the [codebook that describes the variables](#). The final pages of the codebook have information on how to translate variable names to questions in the survey.

1) Data Set creation (23 pts - 3 pts each section, except part 1.5 which is 5 pts)

1.1) Read the data into an object called `envdat` (do NOT use the option `as.is = TRUE`). Check the dimension to be sure the data loaded correctly. Then create a new object called `envdat2` which only contains information for the following countries: Austria, Iceland, Japan, New Zealand, Philippines, Russia, and Thailand. The variable that contains country is `country`. You'll need to use the ISO 3166 Standard Book of Country Codes which you can find [HERE](#).

Check the dimensions of your results - you should have 9476 observations.

```
envdat <-  
read.csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/ISS_2020.csv")  
  
dim(envdat) #21718 rows 224 columns - all loaded correctly  
## [1] 21718 224  
  
# names(envdat)  
  
# subset only wanted countries based on ISO code  
#Austria - 040, "Iceland - 352", "Japan - 392", "New Zealand - 554",  
"Philippines - 608", "Russia - 643", "Thailand - 764"  
envdat2 <- envdat[envdat$country %in% c("40", "352", "392", "554", "608",  
"643", "764"), ]  
  
dim(envdat2) #9476 row 224 variables - correct  
## [1] 9476 224
```

1.2) Create a new variable called Country on envdat2 which has Country names rather than Country numbers. There are several ways to do this, but I suggest you use the `recode()` function in the `car` package. The syntax for this function is something like

```
# recode country to be name
# Austria - 040", "Iceland - 352", "Japan - 392", "New Zealand - 554",
# "Philippines - 608", "Russia - 643", "Thailand - 764"
library(car)
NewVar <- recode(envdat2$country, "040 = 'Austria';
352 = 'Iceland';
392 = 'Japan';
554 = 'New Zealand';
608 = 'Philippines';
643 = 'Russia';
764 = 'Thailand'")
# NAs coerced
```

Once you've created the variable, make a table of the resulting variable to see how many observations there are from each country.

```
# table of new variable
table(NewVar)
```

## NewVar						
##	Austria	Iceland	Japan	New Zealand	Philippines	Russia
##	1261	1150	1491	993	1500	1583
##	Thailand					
##	1498					

1.3) Make a variable Gender on envdat2 that contains gender (which is variable SEX). Recode so that 1 becomes 'Male' and 2 becomes 'Female'. Again, make a table of resulting variable to see how many people identify as Male and how many as Female.

```
# variable of Gender envdat2; 1=Male 2=Female
Gender <- recode(envdat2$SEX,
"1 = 'Male';
2 = 'Female'")
#NAs introduced

# make table of male and female
table(Gender)
```

## Gender	
## Female	Male
##	5032 4444

1.4) Create Variables AgeYears and Educ on envdat2 from variables AGE and EDUCYRS which are age in years and years of education, respectively. Get summary information and make a histogram for each variable to see if values seem reasonable.

Modify both variables so that any negative values are changed to NA. You'll also discover that some people have more than 30 years of education. Assume this is an error and replace these values with NA. Repeat your histograms to make sure your code works.

```
# create AgeYears and Educ
AgeYears <- envdat2$AGE
Educ <- envdat2$EDUCYRS

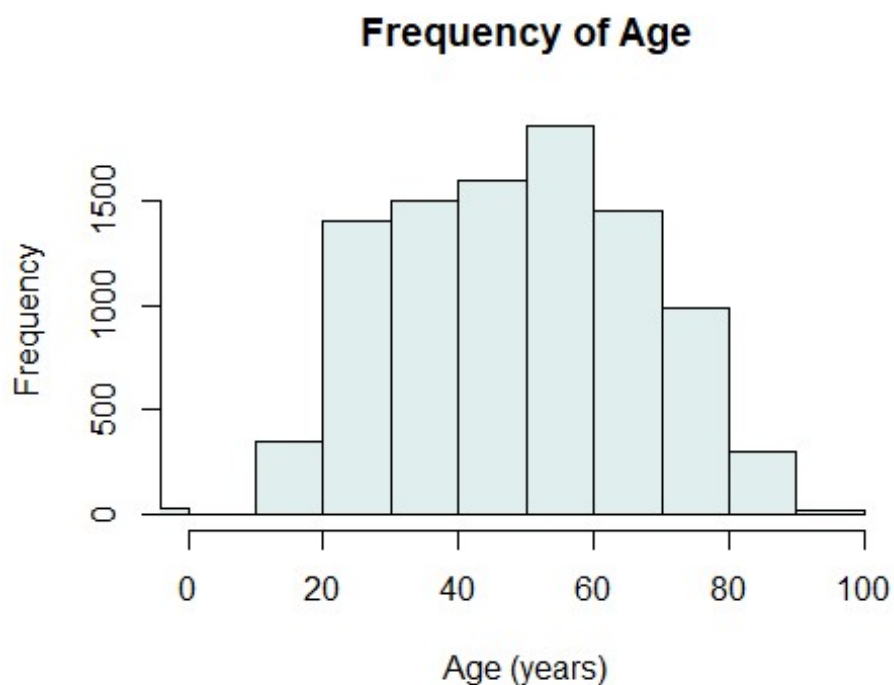
# summary for each
summary(AgeYears)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    -9.00   35.00   50.00   49.35   63.00   97.00

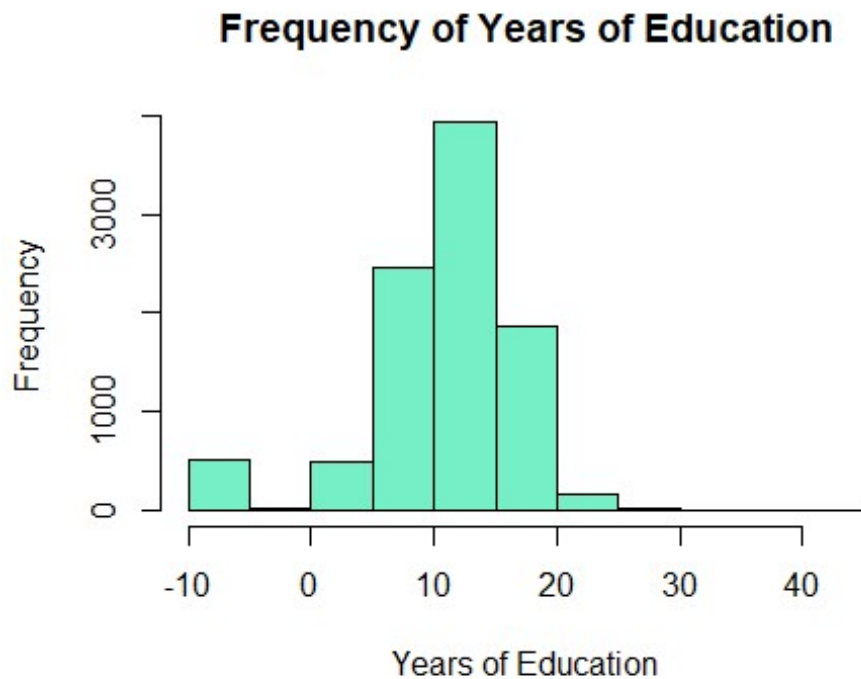
summary(Educ)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    -9.00   10.00   12.00   11.22   15.00   45.00

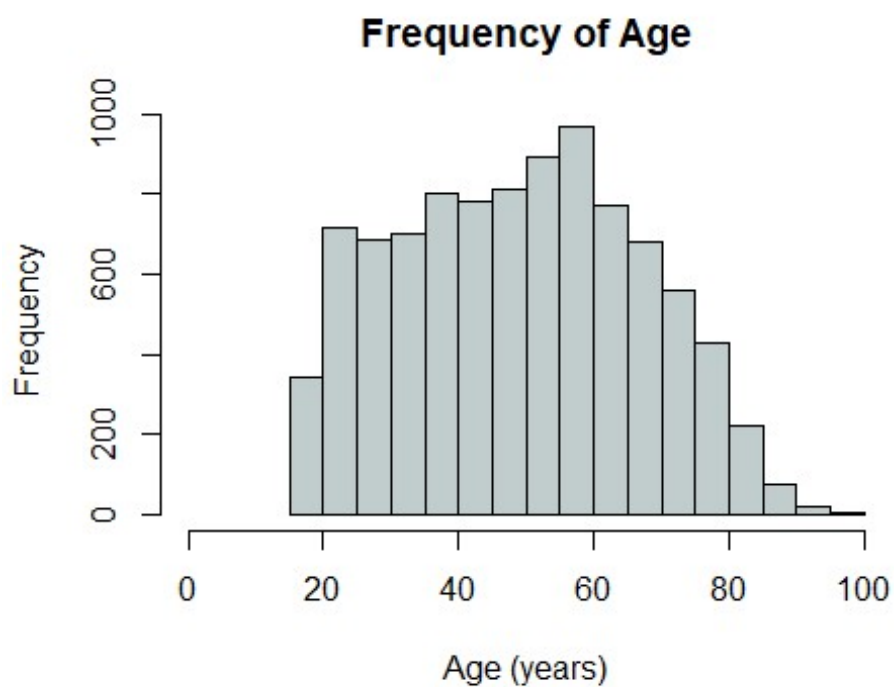
# histogram for each
hist(AgeYears,
     col = "azure2",
     main = "Frequency of Age",
     xlab = "Age (years)",
     xlim = c(0, 100))
```



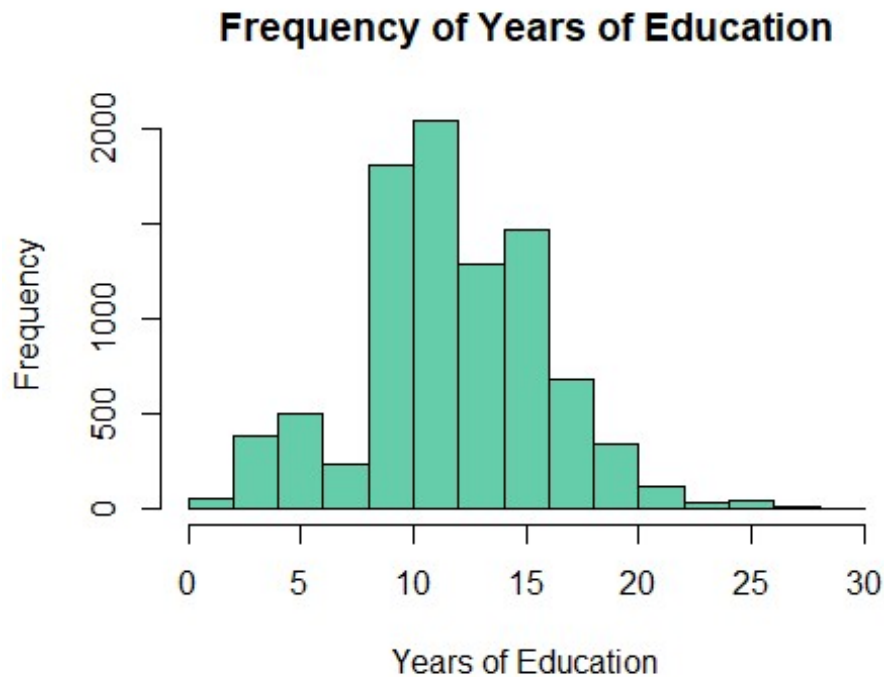
```
hist(Educ,  
     col = "aquamarine2",  
     main = "Frequency of Years of Education",  
     xlab = "Years of Education")
```



```
# any negative values in either variable are changed to NA  
AgeYears[AgeYears < 0] <- NA  
Educ[Educ < 0] <- NA  
  
# if more than 30 years of education, replace with NA  
Educ[Educ > 30] <- NA  
  
# repeat histograms  
hist(AgeYears,  
     col = "azure3",  
     main = "Frequency of Age",  
     xlab = "Age (years)",  
     xlim = c(0, 100))
```



```
hist(Educ,  
     col = "aquamarine3",  
     main = "Frequency of Years of Education",  
     xlab = "Years of Education")
```



1.5) Make a variable `EmpStat` on `envdat2` from variable `MAINSTAT` that contains employment status information. The codebook for this question is listed below:

-9 = No answer -8 = Don't know 1 = In paid work 2 = Unemployed and looking for a job 3 = In education 4 = Apprentice or trainee 5 = Permanently sick or disabled 6 = Retired 7 = Domestic work (meaning working in the home with family) 8 = Incompulsory military service or community service 9 = Other

Code `EmpStat` in the following way : 1, 2, and 8 code as 'Working or Looking', 7 codes as 'At Home', 3 and 4 code as 'Student', 6 codes as 'Retired', -9 and -8 code as NA, and 5, 9 code as 'Other'.

Make a table of your results - look up options for the `table()` function so that it also displays NA values.

```
# EmpStat from MAINSTAT of employment status info
EmpStat <- envdat2$MAINSTAT

# recode EmpStat
EmpStat <- recode(EmpStat,
  "1 = 'Working or Looking';
  2 = 'Working or Looking';
  8 = 'Working or Looking';
  7 = 'At Home';
  3 = 'Student';
  4 = 'Student';
```

```

6 = 'Retired';
-9 = NA;
-8 = NA;
5 = 'Other';
9 = 'Other')
# NA introduced

# make table, including NA
table(EmpStat, useNA = c("always"))

## EmpStat
##           At Home           Other           Retired
Student
##           919           224           1557
437
## Working or Looking      <NA>
##           5983           356

```

1.6) If you look in the codebook starting on page 10, you'll notice many questions are on a 5 point scale. We're going to create a composite environmental index score from 12 of these questions. The new variable should be called EnvAtt on envdat2. It might be tempting to simply add scores together; however, for some questions 'Agree Strongly' (which is coded as 1) would indicate support the environment, while other questions 'Agree Strongly' indicates lack of support for the environment. Furthermore, you want to combined variables so that a higher score indicates more support for the environment.

The following questions should be included : V20, V21, V22, V23; V26, V27, V28, v29; V31, V32, V34, V36. For each question, you'll either want to add it directly, OR you'll want to add (6 - question). Using (6 - question) will convert 1's to 5's and 5's to 1's. Remember, for your final composite variable, you want a HIGHER score to indicate GREATER support for the environment.

FIRST - you'll need to replace the values -9 and -8 in each of these questions (which stand for 'No Answer' and 'Can't Choose') with NA. THEN you can proceed with making your composite variable.

Your new variable should have a minimum of 12 and a maximum of 60. Check this by getting summary information and by making a histogram of EnvAtt. Also, make a normal quantile plot of `Envatt`. You'll also want to determine how many missing values were created in the final index.

The original variables were all on a 5 point scale and were certainly NOT normally distributed. Does your new composite variable seem to be approximately normally distributed?

```

# V20 - Q10a, V21 - Q10b, V22 - Q10c, V23 - Q10d; V26 - Q11a, V27 - Q11b, V28
- Q11c, v29 - Q11d; V31 - Q12b, V32 - Q12c, V34 - Q12e, V36 - Q12g

```

```
# For each question, you'll either want to add it directly, OR you'll want to
add (6 - question). Using (6 - question) will convert 1's to 5's and 5's to
1's
```

```
# Add these directly 5=strongly agree: V20, V21, V23, V29, V32, V34
```

```
# Add 6-question 1=strongly agree: V22, V26, V27, V28, V31, V36
```

```
# replace -9 and -8 with NA
```

```
vars <- c("v20", "v21", "v22", "v23", "v26", "v27", "v28", "v29", "v31",
"v32", "v34", "v36")
```

```
envdat2[vars] <- lapply(envdat2[vars], function(x) ifelse(x %in% c(-9, -8),
NA, x))
```

```
# create new variable `EnvAtt` on `envdat2
```

```
EnvAtt <- envdat2$v20 + envdat2$v21 + envdat2$v23 + envdat2$v29 + envdat2$v32
+ envdat2$v34 + (6-envdat2$v22) + (6-envdat2$v26) + (6-envdat2$v27) + (6-
envdat2$v28) + (6-envdat2$v31) + (6-envdat2$v36)
```

```
# check summary - want min 12, max 60
```

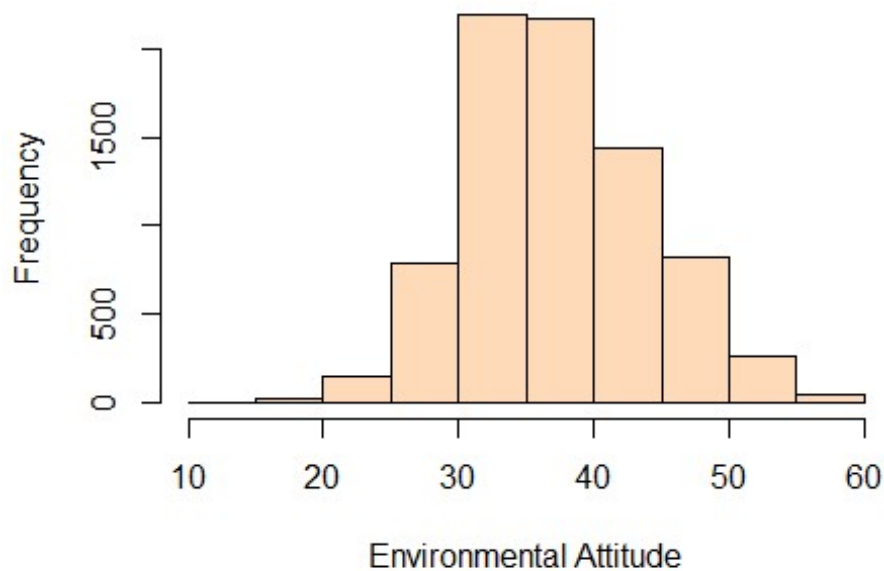
```
summary(EnvAtt)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##      12.00   33.00   37.00   37.83   42.00   60.00     1584
```

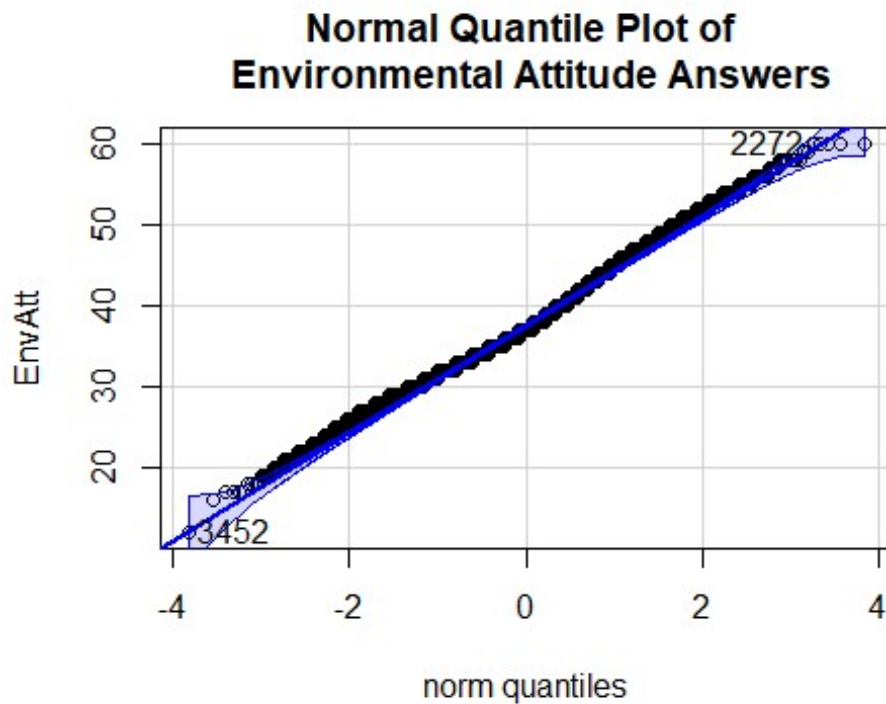
```
# make histogram
```

```
hist(EnvAtt,
      main = "Frequency of Environmental Attitude Answers",
      col = "peachpuff",
      xlab = "Environmental Attitude")
```


Frequency of Environmental Attitude Answers



```
# make normal quantile plot  
qqPlot(EnvAtt,  
        main = "Normal Quantile Plot of \n Environmental Attitude Answers")
```



```
## [1] 3452 2272
```

```
# how many NA in final variable
```

```
sum(is.na(EnvAtt))
```

```
## [1] 1584
```

1584 missing values were created in the final index The new composite variable is normally distributed

1.7) Finally, create a new dataframe called envdat3 which contains only the new variables you've created in 1.2 through 1.6. Remove any rows with missing values for any of these variables. Get the dimension of envdat3. Lastly, attach envdat3.

```
# new df envdat with only new variables of 1.2 - 1.6.
```

```
envdat3 <- data.frame(NewVar, Gender, AgeYears, Educ, EmpStat, EnvAtt)
```

```
# Remove rows with missing values
```

```
envdat3 <- na.omit(envdat3)
```

```
# dimensions
```

```
dim(envdat3) # 7394, 6
```

```
## [1] 7394    6
```

```
# attach
```

```
rm(EnvAtt, AgeYears, Educ, EmpStat, Gender, NewVar)
```

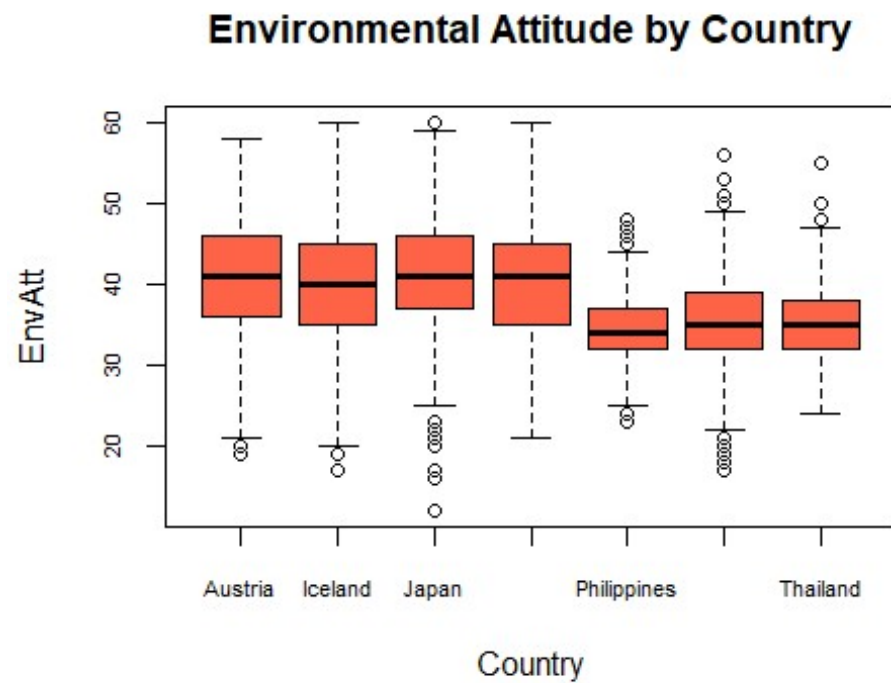
```
attach(envdat3)
```

2. Plots and Interactions (17 pts)

2.1) Make boxplots of EnvAtt by Country, Employment Status, and Gender (that's 3 different boxplots). Additionally, make scatterplots of EnvAtt by age and by years of education. Is there evidence visually that our composite environmental index is related to any of these 5 variables (and elaborate)?

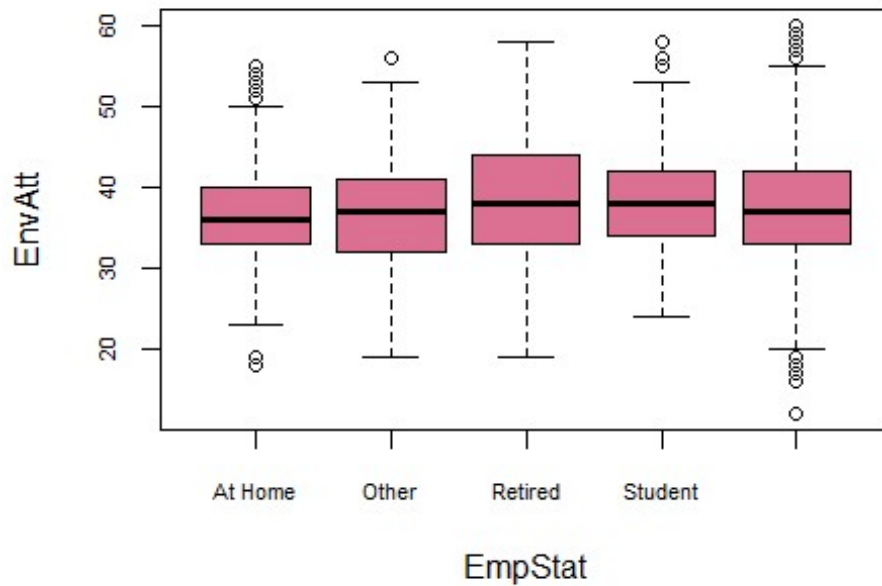
```
# make 3 boxplots of Country, Employment Status, Gender
```

```
boxplot(EnvAtt ~ NewVar,  
        main = "Environmental Attitude by Country",  
        xlab = "Country",  
        col = "tomato",  
        cex.axis = .7)
```



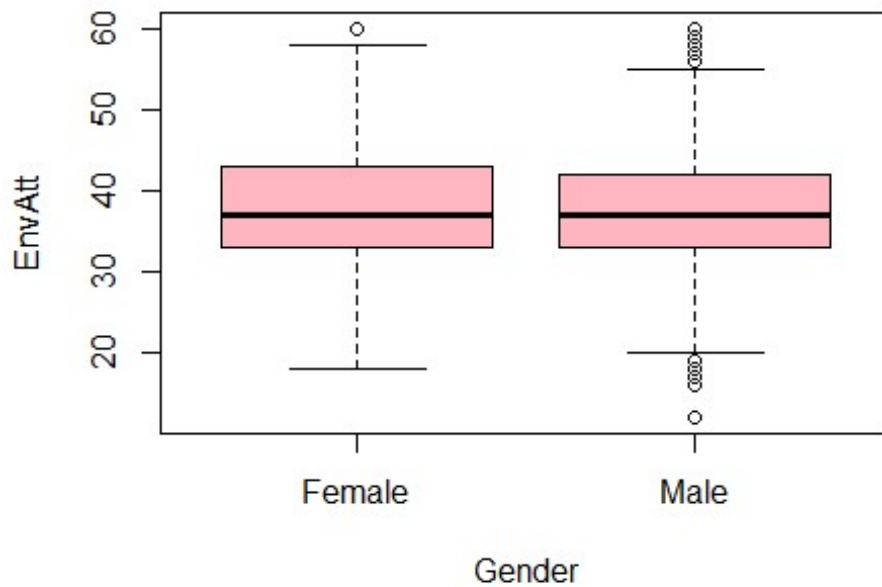
```
boxplot(EnvAtt ~ EmpStat,  
  main = "Environmental Attitude by Employment Status",  
  col = "palevioletred",  
  cex.axis = .7)
```

Environmental Attitude by Employment Status



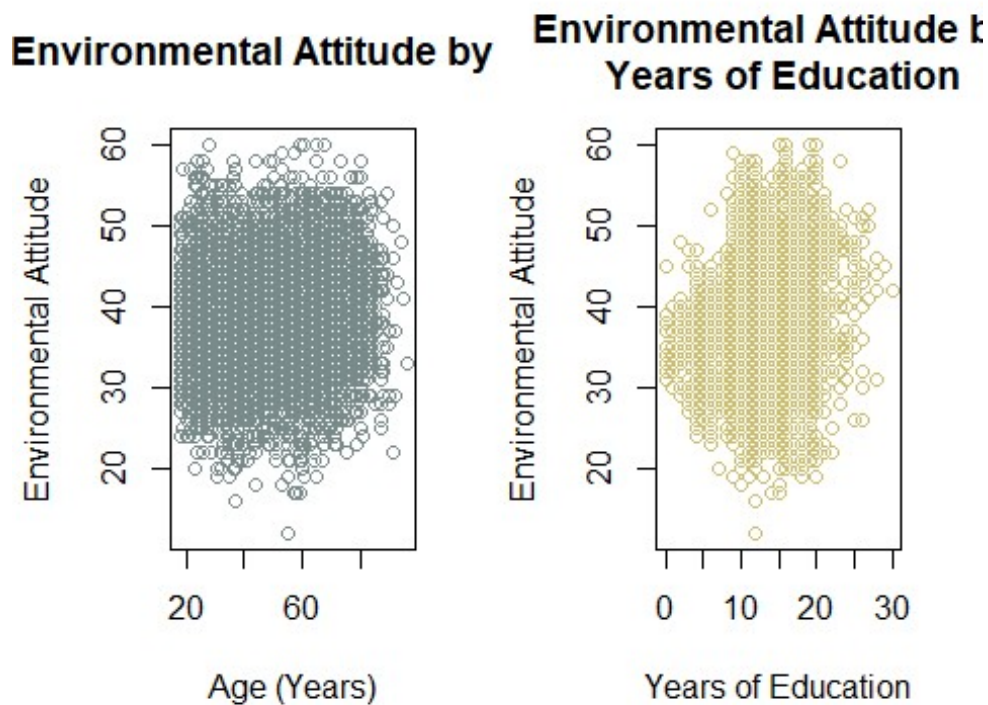
```
boxplot(EnvAtt ~ Gender,  
        main = "Environmental Attitude by Gender",  
        col = "lightpink")
```

Environmental Attitude by Gender



```
# make scatterplots of EnvAtt by age
par(mfrow = c(1, 2))
plot(AgeYears, EnvAtt,
     col = "lightcyan4",
     main = "Environmental Attitude by Age",
     xlab = "Age (Years)",
     ylab = "Environmental Attitude")

# make scatterplots of EnvAtt by years of education
plot(Educ, EnvAtt,
     col = "lightgoldenrod3",
     main = "Environmental Attitude by \n Years of Education",
     xlab = "Years of Education",
     ylab = "Environmental Attitude")
```



Visually, I don't see any evidence that our variables are related to environmental attitude. For the difference between countries, the ranges and medians are all very similar, with Philippines, Russia, and Thailand being slightly lower and tighter spread. For difference between employment status, the medians and spreads are again very similar with some outliers. And for gender, the boxplots are practically equal. For age and years of education, the scatterplots look like blobs without any appreciable trends.

2.2). Examine relationships between the three possible pairs of the variables Gender, Country, EmpStat as they relate to EnvAtt. For each pair of categorical variables, make a table showing counts (i.e. `table(Gender, Country)` for example). This will allow you to see how many individuals exist for each combination of levels of your categorical variables.

THEN, make interaction plots for each pair of variables as they relate to EnvAtt. Which pairs of categorical variables seem like they might have an interaction as they relate to EnvAtt?

```
# relationships between Gender - EnvAtt; Country - EnvAtt; EmpStat - EnvAtt
# make categorical tables: Gender - Country; Country - EmpStat; EmpStat - Gender
```

```
table(Gender, NewVar)
```

```
##           NewVar
## Gender  Austria Iceland Japan New Zealand Philippines Russia Thailand
## Female    544     393   471         417           688     725     583
## Male      555     362   486         390           678     606     496
```

```
table(EmpStat, NewVar)
```

```
##           NewVar
## EmpStat  Austria Iceland Japan New Zealand Philippines Russia
## At Home      22      9   151         30           303    113
## Other        27     45    31          4            21     32
## Retired     405     69   116        199           81    255
## Student      35     45    44         44            82     52
## Working or Looking 610   587   615        530          879    879
```

```
##           NewVar
## EmpStat  Thailand
## At Home      78
## Other         5
## Retired       65
## Student       72
## Working or Looking 859
```

```
table(Gender, EmpStat)
```

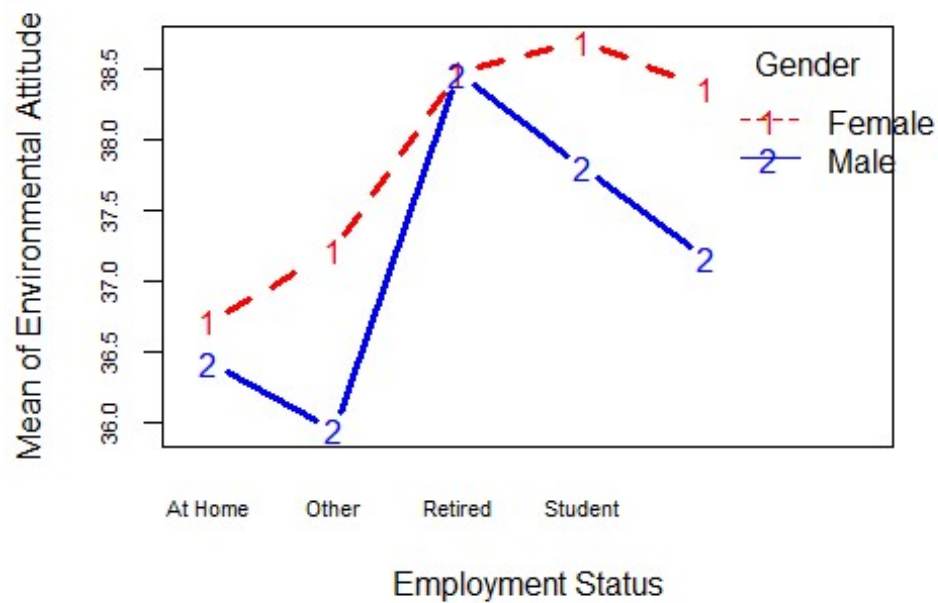
```
##           EmpStat
## Gender  At Home Other Retired Student Working or Looking
## Female    615     88   590     190           2338
## Male       91     77   600     184           2621
```

```
# make interaction plots for each pair of variables as relates to EnvAtt:
Gender - Country - EnvAtt; Country - EmpStat - EnvAtt; EmpStat - Gender - EnvAtt
```

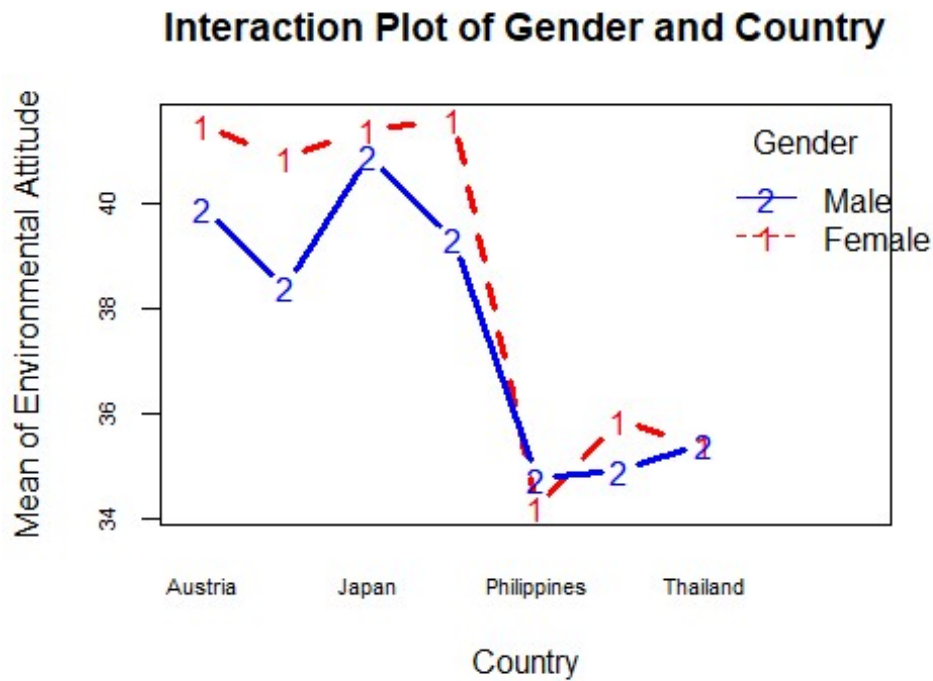
```
interaction.plot(EmpStat, Gender, EnvAtt, type = "b", lwd = 3, col =
c('red','blue','black'), main = "Interaction Plot of Gender and Employment
Status",
```

```
      cex.axis = .7,
      ylab = "Mean of Environmental Attitude",
      xlab = "Employment Status")
```

Interaction Plot of Gender and Employment Statu

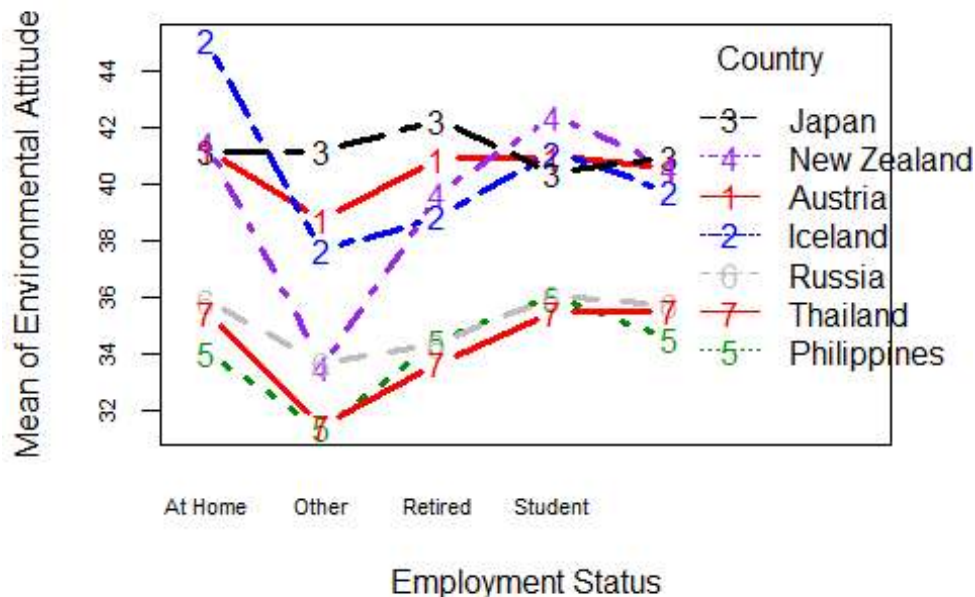


```
interaction.plot(NewVar, Gender, EnvAtt, type = "b", lwd = 3, col =  
c('red', 'blue', 'black'), main = "Interaction Plot of Gender and Country",  
cex.axis = .7,  
ylab = "Mean of Environmental Attitude",  
xlab = "Country")
```



```
interaction.plot(EmpStat, NewVar, EnvAtt, type = "b", lwd = 3, col =
c('red', 'blue', 'black', 'purple', 'green4', 'gray'), main = "Interaction Plot
of Country and Employment Status",
               cex.axis = .7,
               ylab = "Mean of Environmental Attitude",
               xlab = "Employment Status",
               trace.label = "Country")
```


Interaction Plot of Country and Employment Status



All pairs seem to have an interaction as they overlap and do not run parallel. For employment status + gender, they are pretty parallel but cross at retired. For country + gender, they are pretty parallel but cross at Philippines and Thailand. For country + employment status, Philippines, Thailand, and Russia run almost parallel but also on top of each other at some points (suggesting multicollinearity) and Austria, New Zealand, Iceland, and Japan also run parallel but cross in some areas. Japan also crosses with Russia. This suggests that there could be interaction across many variables.

3. Fitting Two-Way ANOVA models (20 pts, 10 pts each part)

3.1) Fit three different two-way ANOVA models for EnvAtt, one for each pair of your three categorical variables (i.e. Country and EmpStat, Country and Gender, Gender and EmpStat). Include an interaction term in each model. Get summary information for each model using the `Anova()` function with option `type = 'III'`. Which models appear to have significant interaction terms? Does this seem to be consistent with what you observed in the interaction plots above?

```
library(car)
# fit 3 two-way ANOVA for EnvAtt; Country and EmpStat, Country and Gender,
# Gender and EmpStat with interaction term
# create model for each interaction
lm1 <- lm(EnvAtt ~ NewVar + EmpStat + NewVar*EmpStat)
lm2 <- lm(EnvAtt ~ NewVar + Gender + NewVar*Gender)
lm3 <- lm(EnvAtt ~ Gender + EmpStat + Gender*EmpStat)
```

```

# summery of each model with Anova(, type = 3)
Anova(lm1, type = 'III')

## Anova Table (Type III tests)
##
## Response: EnvAtt
##           Sum Sq   Df    F value    Pr(>F)
## (Intercept)  37393    1 1033.5723 < 2.2e-16 ***
## NewVar       7217     6   33.2475 < 2.2e-16 ***
## EmpStat      136      4    0.9425  0.438072
## NewVar:EmpStat 1719   24    1.9803  0.002955 **
## Residuals    266238 7359
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Anova(lm2, type = 'III')

## Anova Table (Type III tests)
##
## Response: EnvAtt
##           Sum Sq   Df    F value    Pr(>F)
## (Intercept) 935991    1 26007.7321 < 2.2e-16 ***
## NewVar      37965     6   175.8169 < 2.2e-16 ***
## Gender       686      1    19.0727 1.275e-05 ***
## NewVar:Gender 1980     6     9.1676 5.048e-10 ***
## Residuals    265599 7380
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Anova(lm3, type = 'III')

## Anova Table (Type III tests)
##
## Response: EnvAtt
##           Sum Sq   Df    F value    Pr(>F)
## (Intercept) 829402    1 18981.5333 < 2.2e-16 ***
## Gender        7       1     0.1579  0.69112
## EmpStat      1545     4     8.8410 4.06e-07 ***
## Gender:EmpStat 374     4     2.1396  0.07325 .
## Residuals    322645 7384
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

The interaction between Country and Employment status, Country and the interaction term is significant but employment status is not.

The interaction between Country and Gender, all terms are significant.

The interaction between Gender and Employment status, Gender is not significant but employment status and the interaction is at alpha of 0.1.

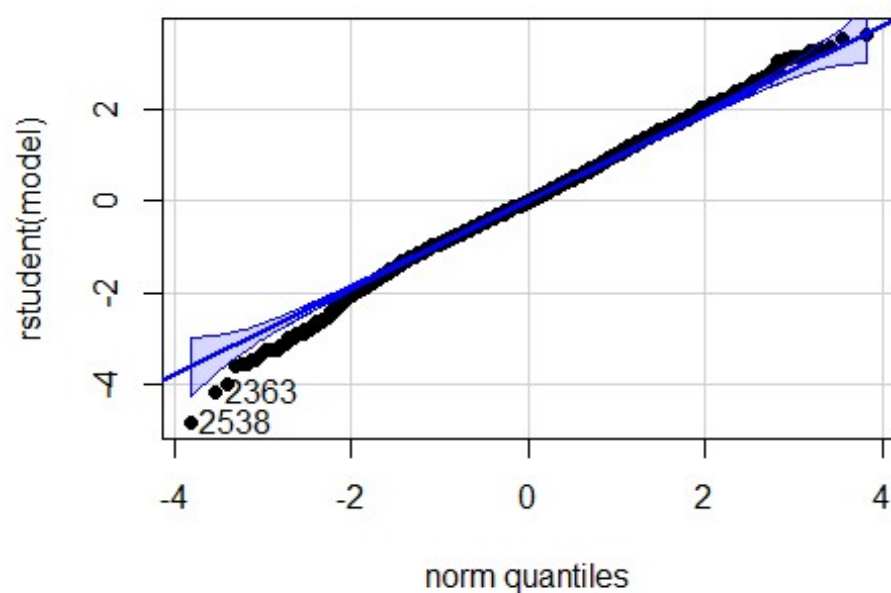
Yes, this is pretty consistent with what I saw in the interaction plots with most having

interactions and Country and Employment status or Gender having more interaction than Gender and Employment Status.

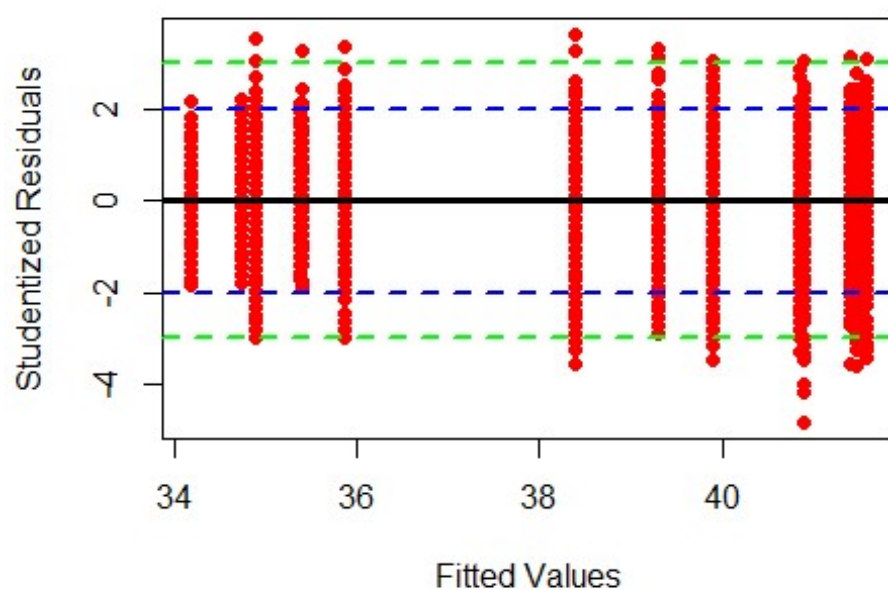
3.2) For the model with Country and Gender you fit in part (3.1), make residual plots. Are the model assumptions reasonably met (write a sentence supporting your answer)?

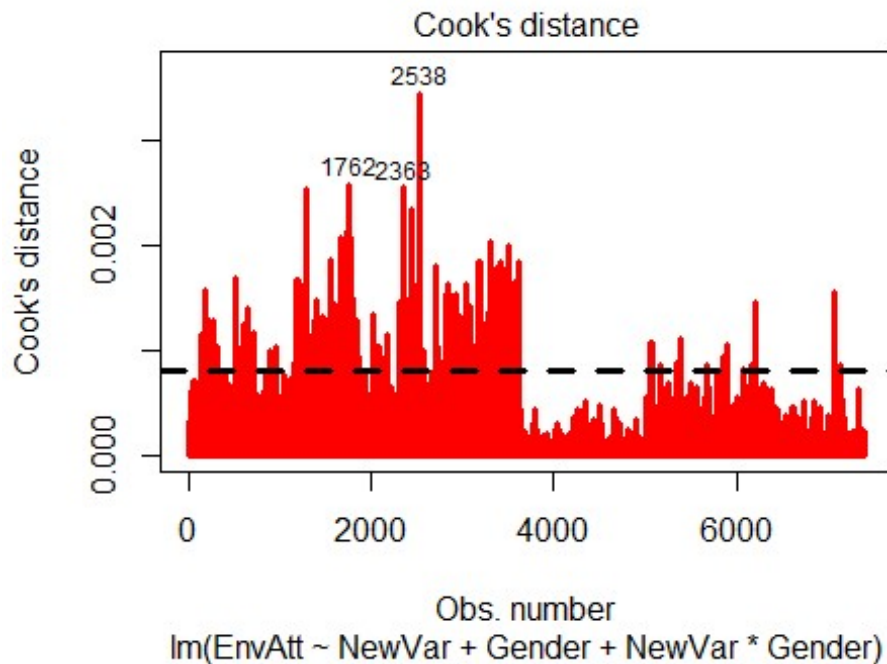
```
# residual plot for model with country and gender
download.file("https://raw.githubusercontent.com/jreuning/sds230_files/refs/heads/main/regJDRS.txt",
              destfile = "regJDRS.txt")
source("regJDRS.txt")
myResPlots(lm2, "Country and Gender")
```

NQ Plot of Studentized Residuals, Country and Gen



Fits vs. Studentized Residuals, Country and Gen





Yes, the model assumptions are met - the errors are reasonably normal, there is not evidence of heteroskedasticity or largely influential points.

4. ANCOVA (20 pts, 10 pts each part)

4.1) Fit an ANCOVA model predicting EnvAtt based on Gender, Education, and the interaction of Gender and Education. Get ANOVA summary information for this model (again, use the `Anova()` function with option `type = 'III'`). Is there a significant interaction between Gender and Education? Also get linear model summary information for this model. For which gender is there a greater increase in Environmental Index Score as Education increases?

```
# fit ANCOVA model predicting EnvAtt based on Gender, education and interaction
```

```
lm4 <- lm(EnvAtt ~ Gender + Educ + Educ * Gender)
Anova(lm4, type = 3)
```

```
## Anova Table (Type III tests)
```

```
##
```

```
## Response: EnvAtt
```

	Sum Sq	Df	F value	Pr(>F)	
(Intercept)	350402	1	8508.628	< 2.2e-16	***
Gender	604	1	14.661	0.0001298	***
Educ	15574	1	378.170	< 2.2e-16	***
Gender:Educ	1194	1	28.996	7.475e-08	***
Residuals	304335	7390			

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# Linear model summary
summary(lm4)

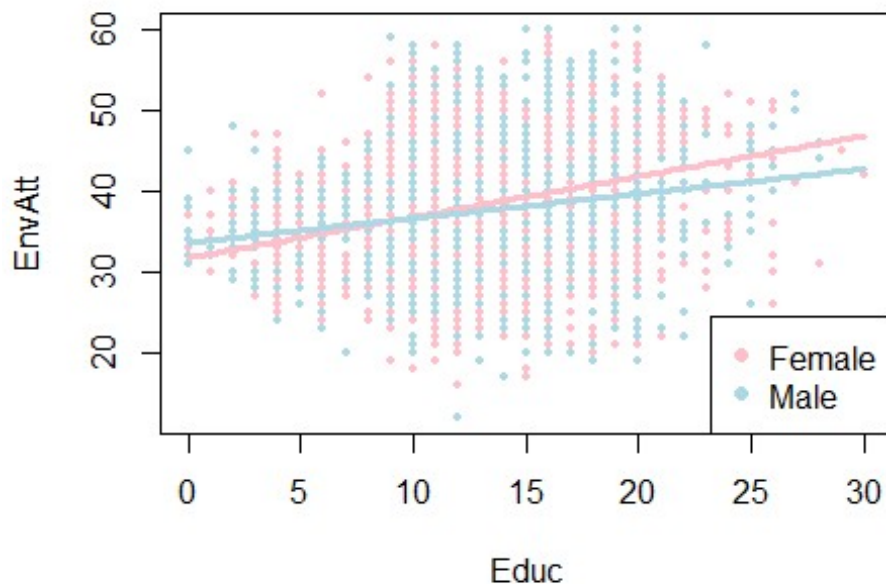
##
## Call:
## lm(formula = EnvAtt ~ Gender + Educ + Educ * Gender)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -25.2458  -4.2849  -0.4226   4.2188  22.6658
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   31.72925    0.34398   92.242  < 2e-16 ***
## GenderMale     1.87018    0.48843    3.829  0.00013 ***
## Educ           0.50371    0.02590   19.447  < 2e-16 ***
## GenderMale:Educ -0.19984    0.03711   -5.385  7.48e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.417 on 7390 degrees of freedom
## Multiple R-squared:  0.0673, Adjusted R-squared:  0.06692
## F-statistic: 177.8 on 3 and 7390 DF,  p-value: < 2.2e-16
```

Yes, all terms are significant in the ANOVA including the interaction between Gender and Education. From the linear summary, as education increases, there is a greater increase of environmental attitude in women.

4.2) Make a plot that shows EnvAtt as predicted by years of education with separate colors for each gender. Then, superimpose the two predicted regression lines (one for each gender). The plot should be consistent with your results from part a). You'll want to look carefully at similar code in Class 19.

```
mod1 <- lm(EnvAtt ~ Educ * Gender)
plot(EnvAtt ~ Educ, col = c("pink", "lightblue"), pch = 16, cex = .5)
legend("bottomright", col = c("pink", "lightblue"), legend =
levels(factor(Gender)), pch = 16)

Conlevs <- levels(factor(Gender))
for (i in 1:2){
  curve(predict(mod1, newdata = data.frame(Educ = x, Gender = Conlevs[i])),
col = c("pink", "lightblue")[i], lwd = 3, add=TRUE)
}
```



5. GLM (20 pts, 10 pts each part)

5.1). Fit a model for EnvAtt that includes ALL of the five possible continuous and categorical predictors. Also include two-way interactions between Gender and Education, Employment Status and Country, Gender and Country, Employment Status and Gender, Age and Gender. Save to an object called m1.

Get ANOVA summary information for this model (again, use `Anova()` with option `type = 'III'`).

THEN, perform manual backwards stepwise regression, removing non-significant terms until all terms have p-values less than 0.05. REMEMBER, you want to remove interactions BEFORE you remove any main effects. You don't need to show every step - just put your final model into an object called m2. Get linear model summary information for this model. Finally, check residuals for your final model.

```
# fit model for EnvAtt including all five continuous and categorical
# predictors including two way interactions
names(envdat3)

## [1] "NewVar"    "Gender"    "AgeYears"  "Educ"      "EmpStat"   "EnvAtt"

m1 <- lm(EnvAtt ~ Gender + AgeYears + Educ + NewVar + EmpStat + Gender*Educ +
EmpStat*NewVar + Gender*NewVar + EmpStat*Gender + AgeYears*Gender)
```

```

# get ANOVA
Anova(m1, type = 3)

## Anova Table (Type III tests)
##
## Response: EnvAtt
##           Sum Sq   Df F value    Pr(>F)
## (Intercept) 23211    1 663.2414 < 2.2e-16 ***
## Gender        3      1  0.0875 0.7674492
## AgeYears     70      1  2.0065 0.1566664
## Educ        4235     1 121.0026 < 2.2e-16 ***
## NewVar      4452     6  21.2049 < 2.2e-16 ***
## EmpStat     177      4   1.2651 0.2812535
## Gender:Educ   417     1  11.9180 0.0005591 ***
## NewVar:EmpStat 1452    24   1.7289 0.0149056 *
## Gender:NewVar   913     6   4.3503 0.0002168 ***
## Gender:EmpStat   85     4   0.6078 0.6569962
## Gender:AgeYears    7     1   0.1920 0.6612754
## Residuals    257008 7344
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# backwards stepwise regression, until pvalues < .5
# gender and age; gender and employment
m2 <- lm(EnvAtt ~ Gender + Educ + NewVar + EmpStat + Gender*Educ +
EmpStat*NewVar + Gender*NewVar)

# Linear model summary
Anova(m2, type = 3)

## Anova Table (Type III tests)
##
## Response: EnvAtt
##           Sum Sq   Df F value    Pr(>F)
## (Intercept) 29487    1 842.6723 < 2.2e-16 ***
## Gender        2      1  0.0455 0.8311805
## Educ        4483     1 128.1198 < 2.2e-16 ***
## NewVar      5242     6  24.9670 < 2.2e-16 ***
## EmpStat     188      4   1.3414 0.2519121
## Gender:Educ   494     1  14.1152 0.0001733 ***
## NewVar:EmpStat 1554    24   1.8500 0.0069771 **
## Gender:NewVar   898     6   4.2783 0.0002606 ***
## Residuals    257190 7350
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(m2)

##
## Call:
## lm(formula = EnvAtt ~ Gender + Educ + NewVar + EmpStat + Gender *

```



```

##      Educ + EmpStat * NewVar + Gender * NewVar)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -28.2838  -3.7339  -0.0315   3.7725  20.9013
##
## Coefficients:
##                                Estimate Std. Error t value
## (Intercept)                   37.706926   1.298948   29.029
## GenderMale                     0.127674   0.598859    0.213
## Educ                           0.329564   0.029116   11.319
## NewVarIceland                  3.300957   2.355106    1.402
## NewVarJapan                   -0.605495   1.351492   -0.448
## NewVarNew Zealand             -0.907834   1.667581   -0.544
## NewVarPhilippines             -6.735039   1.307686   -5.150
## NewVarRussia                  -5.963427   1.382558   -4.313
## NewVarThailand                -5.217981   1.429540   -3.650
## EmpStatOther                  -2.323669   1.701006   -1.366
## EmpStatRetired                 0.226034   1.306504    0.173
## EmpStatStudent                 0.474699   1.630802    0.291
## EmpStatWorking or Looking     -0.167041   1.299529   -0.129
## GenderMale:Educ               -0.151873   0.040424   -3.757
## NewVarIceland:EmpStatOther    -4.955253   2.749852   -1.802
## NewVarJapan:EmpStatOther      2.654208   2.073359    1.280
## NewVarNew Zealand:EmpStatOther -3.584369   3.596212   -0.997
## NewVarPhilippines:EmpStatOther -0.626657   2.172746   -0.288
## NewVarRussia:EmpStatOther     0.303754   2.073750    0.146
## NewVarThailand:EmpStatOther   -1.105699   3.221452   -0.343
## NewVarIceland:EmpStatRetired  -5.965260   2.471088   -2.414
## NewVarJapan:EmpStatRetired    1.195620   1.524454    0.784
## NewVarNew Zealand:EmpStatRetired -0.528991   1.755272   -0.301
## NewVarPhilippines:EmpStatRetired -0.189294   1.515979   -0.125
## NewVarRussia:EmpStatRetired   -1.614991   1.467928   -1.100
## NewVarThailand:EmpStatRetired  -1.696195   1.651969   -1.027
## NewVarIceland:EmpStatStudent  -5.111654   2.707488   -1.888
## NewVarJapan:EmpStatStudent    -1.142221   1.930822   -0.592
## NewVarNew Zealand:EmpStatStudent 1.126119   2.155986    0.522
## NewVarPhilippines:EmpStatStudent 0.376934   1.794465    0.210
## NewVarRussia:EmpStatStudent   0.062844   1.909744    0.033
## NewVarThailand:EmpStatStudent  -2.014445   1.901588   -1.059
## NewVarIceland:EmpStatWorking or Looking -5.551059   2.374535   -2.338
## NewVarJapan:EmpStatWorking or Looking -0.136706   1.420224   -0.096
## NewVarNew Zealand:EmpStatWorking or Looking -0.079605   1.713897   -0.046
## NewVarPhilippines:EmpStatWorking or Looking 0.146312   1.368005    0.107
## NewVarRussia:EmpStatWorking or Looking 0.192300   1.431128    0.134
## NewVarThailand:EmpStatWorking or Looking -0.536434   1.481178   -0.362
## GenderMale:NewVarIceland      0.127378   0.595785    0.214
## GenderMale:NewVarJapan        1.226117   0.559254    2.192
## GenderMale:NewVarNew Zealand  -0.006406   0.570642   -0.011
## GenderMale:NewVarPhilippines  1.999056   0.512325    3.902

```

```

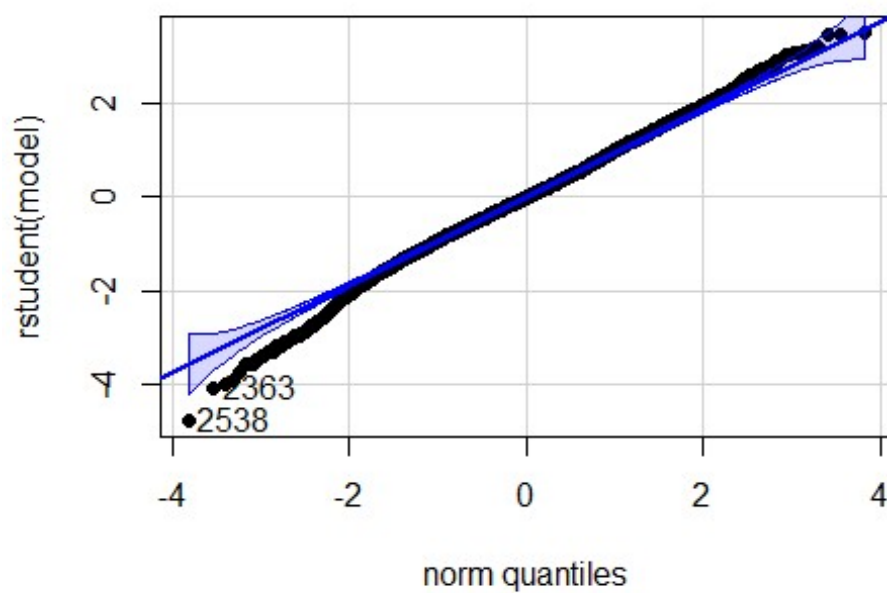
## GenderMale:NewVarRussia          0.849649    0.499080    1.702
## GenderMale:NewVarThailand         1.726795    0.519496    3.324
##                                Pr(>|t|)
## (Intercept)                      < 2e-16 ***
## GenderMale                       0.831181
## Educ                             < 2e-16 ***
## NewVarIceland                    0.161072
## NewVarJapan                      0.654152
## NewVarNew Zealand                0.586181
## NewVarPhilippines                2.67e-07 ***
## NewVarRussia                     1.63e-05 ***
## NewVarThailand                    0.000264 ***
## EmpStatOther                     0.171963
## EmpStatRetired                   0.862651
## EmpStatStudent                   0.770996
## EmpStatWorking or Looking        0.897725
## GenderMale:Educ                  0.000173 ***
## NewVarIceland:EmpStatOther        0.071585 .
## NewVarJapan:EmpStatOther          0.200533
## NewVarNew Zealand:EmpStatOther    0.318940
## NewVarPhilippines:EmpStatOther    0.773036
## NewVarRussia:EmpStatOther         0.883550
## NewVarThailand:EmpStatOther        0.731435
## NewVarIceland:EmpStatRetired       0.015802 *
## NewVarJapan:EmpStatRetired         0.432893
## NewVarNew Zealand:EmpStatRetired   0.763139
## NewVarPhilippines:EmpStatRetired   0.900633
## NewVarRussia:EmpStatRetired        0.271288
## NewVarThailand:EmpStatRetired       0.304562
## NewVarIceland:EmpStatStudent       0.059069 .
## NewVarJapan:EmpStatStudent         0.554155
## NewVarNew Zealand:EmpStatStudent   0.601462
## NewVarPhilippines:EmpStatStudent   0.833632
## NewVarRussia:EmpStatStudent        0.973750
## NewVarThailand:EmpStatStudent       0.289476
## NewVarIceland:EmpStatWorking or Looking 0.019427 *
## NewVarJapan:EmpStatWorking or Looking 0.923319
## NewVarNew Zealand:EmpStatWorking or Looking 0.962955
## NewVarPhilippines:EmpStatWorking or Looking 0.914830
## NewVarRussia:EmpStatWorking or Looking 0.893114
## NewVarThailand:EmpStatWorking or Looking 0.717238
## GenderMale:NewVarIceland          0.830710
## GenderMale:NewVarJapan            0.028381 *
## GenderMale:NewVarNew Zealand      0.991044
## GenderMale:NewVarPhilippines      9.63e-05 ***
## GenderMale:NewVarRussia           0.088717 .
## GenderMale:NewVarThailand          0.000892 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

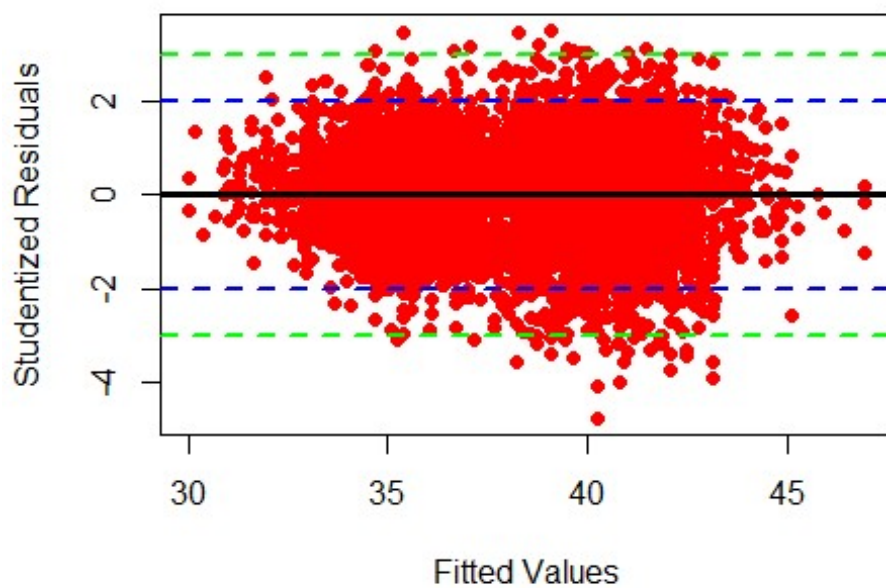
```
## Residual standard error: 5.915 on 7350 degrees of freedom
## Multiple R-squared:  0.2118, Adjusted R-squared:  0.2072
## F-statistic: 45.93 on 43 and 7350 DF,  p-value: < 2.2e-16
```

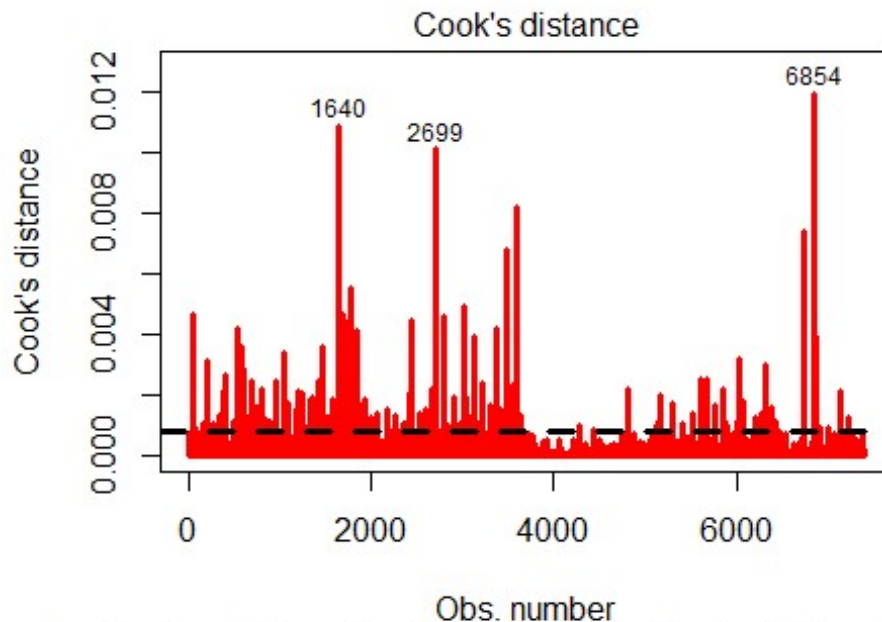
```
# check residuals
myResPlots(m2, "")
```

NQ Plot of Studentized Residuals,



Fits vs. Studentized Residuals,





$\text{nvAtt} \sim \text{Gender} + \text{Educ} + \text{NewVar} + \text{EmpStat} + \text{Gender} * \text{Educ} + \text{EmpStat} * \text{Country}$

5.2). Write a few sentences describing the overall fit of your final model, the direction of coefficients for each continuous predictor, some discussion of categorical predictors, and the nature of any resulting interactions.

The terms kept in the ultimate model were: gender, education, country, employment status, and the interactions between gender and education, country and employment status, and gender and country. All terms that were significant were significant to a 0.01 alpha with only gender and employment status not being significant on their own but needed to be retained for the interaction. This model accounts for about 21% of the variation in environmental attitude scores. The overall p value is practically 0 making it significant at any alpha, indicating the model as a whole meaningfully predicts environmental attitudes. In the linear model, the reference variables were female, Austria, At Home. Education was a continuous variable. As education increased so did environmental attitudes (positive coefficient), although the interaction for males and education was negative, reflecting what we saw in earlier plots that the positive relationship is weaker for males than females. Iceland is the only country that had higher environmental attitudes (all being compared to Austria) though not a statistically significant difference. Employment status on it's own was not significant but it's interaction with country was, meaning relationship between employment status and environmental attitudes vary between countries. Gender and Country interaction was also significant, particularly because males in Japan, Philippines, and Thailand had higher environmental attitudes when compared to females than the reference. From the residual plots we can tell this model does meet all the models assumptions - the errors are normal, there is no heteroskedasticity, and no large influential points.

THE END